Gamete Machine

Following individual alleles through meiosis, and reading a gamete closely enough to say *which division* failed.

Name ____________________ Date ____________________ Period ____________________

Demonstration: andresdbp.github.io/resources/demos/gamete-machine/

1 Before you open anything

Answer from what you already know. The point is to have a number on paper that the demonstration can later confirm or contradict.

PREDICTION

One meiosis produces four gametes. Consider two cells that each make one mistake. In cell X, two homologues fail to separate at **anaphase I**. In cell Y, anaphase I goes perfectly but two sister chromatids fail to separate at **anaphase II**. Out of the four gametes, how many end up with the wrong chromosome number in each cell?

Cell X (anaphase I error): ______ of 4 Cell Y (anaphase II error): ______ of 4

Write one or two sentences saying how you arrived at those two numbers.

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2 Setting the machine up and reading it

Open the demonstration. It models one diploid cell with 2n = 4: chromosome 1 carries genes **A** and **B**, chromosome 2 carries gene **D**. Colour shows which parent a stretch of DNA came from.

1. Confirm the starting settings. Under **Nondisjunction** the **None** tab is selected and the chromosome tab is **Chr 2 (D)**. Under **Crossing over (chromosome 1)** the box **Show a cell that got the crossover** is ticked, **Crossover position** reads **0.50** and **Crossover frequency** reads **40%**. Under **Independent assortment**, **Arrangement 1** is selected. Reload the page if anything differs.
2. Drag the **Stage** slider to **1 of 8**, then press **Next** once. For each of these two stages, copy the stage name printed under the slider and the count line printed along the bottom of the cell picture.

Stage	Stage name under the slider	Count line at the bottom of the cell picture
1 of 8		
2 of 8		

a. Between those two stages one of the two counts doubled and the other did not. Which is which, and why does the cell still count as 2n = 4?

1. Press **Next** until the Stage control reads **8 of 8**, or press **Play** and let it run to the end. Read the four gamete cards under the heading *The four gametes this meiosis made*.
2. Record the letters in each card's **Chr 1** and **Chr 2** rows, and the label in the top-right corner of the card.

Card	Chr 1 letters	Chr 2 letters	Label top-right of the card
Gamete 1			
Gamete 2			
Gamete 3			
Gamete 4			

b. The two parents of this cell supplied only **A B** and **a b** on chromosome 1. Which gamete cards carry a chromosome 1 that neither parent had, and what process produced them?

c. Record these four readouts as they stand now.

Abnormal gametes	n+1 / n-1	Recombination frequency A–B	Recombinant chromatids in this cell

d. The Crossover frequency is set to 40%, but the recombination frequency you just recorded is not 40%. Read the hint under the Crossover frequency slider and explain the relationship between the two numbers.

3 Meiosis I error versus meiosis II error

Leave every other control where it is. Change only the three tabs in the **Nondisjunction** panel, with **Chr 2 (D)** selected as the chromosome that mis-sorts, and stay at Stage **8 of 8**.

1. With the tab on **None**, fill in the first row.
2. Click **Meiosis I** and fill in the second row.
3. Click **Meiosis II** and fill in the third row.
4. For the last column, find a card labelled **n + 1 = 3** and copy the letters sitting in its **Chr 2** row. If there is no such card, write “none”.

Nondisjunction tab	Abnormal gametes	n+1 / n-1	Chr 2 letters in an n + 1 card	Same allele twice, or two different?
None				
Meiosis I				
Meiosis II				

a. State the rule in your own words: what does a meiosis I failure do to the four gametes that a meiosis II failure does not?

b. Go back to your prediction in Part 1. Do the two counts match what the *Abnormal gametes* readout gave? If not, name the assumption behind your prediction that turned out not to hold.

c. Explain the counts mechanistically. Anaphase I happens before the cell has divided at all; anaphase II happens after it has already split in two. Use that to account for 4 of 4 versus 2 of 4.

d. Now account for the alleles. Why does an extra chromosome from a meiosis I failure carry **D and d**, while one from a meiosis II failure carries **D and D**? Name the two structures that each division is supposed to separate.

4 Where the signature stops being reliable

Part 3 gives you a diagnostic: two different alleles on the extra chromosome means meiosis I, the same allele twice means meiosis II. That diagnostic is genuinely used on real human trisomies — and it has a condition attached. Here is where it fails.

1. Set **Nondisjunction** to **Meiosis II** and the chromosome tab to **Chr 1 (A, B)**.
2. Keep **Show a cell that got the crossover** ticked and stay at Stage **8 of 8**.
3. Set **Crossover position** to each value in the table. Gene **A** sits at 0.30 and gene **B** at 0.75, both measured from the centromere. Find the card labelled **n + 1 = 3** and copy the two chips in its **Chr 1** row.

Crossover position	Chr 1 chips in the n + 1 card	Alleles at gene A	Alleles at gene B	Read on its own, does gene B say MI or MII?
0.20				
0.50				
0.80				

a. At position 0.50 the two chromatids in the n + 1 gamete are sisters, yet they do not match at gene B. Explain how two sister chromatids can differ at all.

b. At position 0.20 the crossover lies proximal to both genes. What does the extra chromosome read at A and at B, and why would that data alone be misread as a meiosis I error?

c. Working backwards. Geneticists deciding whether a real trisomy arose at MI or MII genotype markers *near the centromere* and ignore markers near the tip. Using your table, justify that choice in two sentences.

> Your table shows the condition on the diagnostic, not a loophole that ruins it. A crossover landing between the centromere and the marker is uncommon; a crossover landing somewhere out along the arm is not. That asymmetry is the whole reason centromere-proximal markers are trusted.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen the demonstration, set the controls the problem describes, and check yourself — and where you were wrong, write down which step went off.

1. The Crossover frequency slider is moved to 60% and the Crossover position to 0.50. (a) Predict the recombination frequency A–B and the map distance in cM. (b) Predict both again for a Crossover position of 0.85, same 60% frequency. (c) Explain why no setting of the frequency slider can push the recombination frequency above 50%.

2. One meiosis in a cell with 2n = 4 produced these four gametes. Chromosome 1 carries genes A and B; chromosome 2 carries gene D.

	Gamete 1	Gamete 2	Gamete 3	Gamete 4
Chromosome 1	A B	A b	a b	a B
Chromosome 2	D, d	D, d	none	none

(a) Write each gamete's chromosome number as n, n + 1 or n - 1. (b) Which division failed, and what in the chromosome 2 data tells you? (c) Did a crossover occur on chromosome 1, and can you tell whether it fell between A and B? Justify from the letters.

3. A grasshopper has 2n = 8. Ignoring crossing over, how many chromosomally distinct gametes can one individual produce, and how many distinct types appear among the four products of a *single* meiosis? Show the reasoning, not just the numbers.

4. Explain, without computing anything: an n + 1 gamete is produced by a meiosis II error on a chromosome carrying gene A near the centromere and gene B out on the arm. Genotyping gene B suggests a meiosis I error while genotyping gene A does not. State what must have happened during prophase I, and where on the chromosome it must have happened.

6 On your own

Nothing here is graded, and the demonstration does considerably more than this worksheet used.

- Set Nondisjunction back to **None** and go to Stage **4 of 8**. Flip between **Arrangement 1** and **Arrangement 2** and follow chromosome 2 through to Stage 8. The *Combinations from two pairs* readout says 4, but one meiosis shows only 2 of them — work out what the other two look like, and what the *Human, 2^{23}* readout is counting.
- Drag **Crossover frequency** down to 0% and watch what happens to the **Show a cell that got the crossover** tickbox and the hint under it. Why should the single-cell picture not be allowed to show a crossover when no meiosis in the population gets one?
- Put the failure on **Chr 1 (A, B)** instead of Chr 2 and compare the *Meiosis I* gamete cards with the chromosome 2 version from Part 3. What is identical about the two cases, and what is not?
- Read the **Honest limits** panel. Pick the one simplification listed there that you think would change the picture most if it were removed, and say what would change.